

FLPurifier: Backdoor Defense in Federated Learning via Decoupled Contrastive Training

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Abstract—Recent studies have demonstrated that backdoor attacks can cause a significant security threat to federated learning. Existing defense methods mainly focus on detecting or eliminating the backdoor patterns after the model is backdoored. However, these methods either cause model performance degradation or heavily rely on impractical assumptions, such as labeled clean data, which exhibit limited effectiveness in federated learning. To this end, we propose *FLPurifier*, a novel backdoor defense method in federated learning that can effectively purify the possible backdoor attributes before federated aggregation. Specifically, *FLPurifier* splits a complete model into a feature extractor and classifier, in which the extractor is trained in a decoupled contrastive manner to break the strong correlation between trigger features and the target label. Compared with existing backdoor mitigation methods, *FLPurifier* doesn't rely on impractical assumptions since it can effectively purify the backdoor effects in the training process rather than an already trained model. Moreover, to decrease the negative impact of backdoored classifiers and improve global model accuracy, we further design an adaptive classifier aggregation strategy to dynamically adjust the weight coefficients. Extensive experimental evaluations on six benchmark datasets demonstrate that *FLPurifier* is effective

against known backdoor attacks in federated learning with negligible performance degradation and outperforms the state-of-the-art defense methods.

Index Terms—Federated learning, backdoor attacks, decoupled contrastive training, adaptive classifier aggregation.

I. INTRODUCTION

FEDERATED learning (FL) is an emerging collaborative learning paradigm that improves privacy by enabling clients to keep their training data locally. Particularly, it enables machine learning models to be continuously improved and optimized through distributed and localized training [1], [2]. Despite its benefits, FL has been demonstrated to be vulnerable to backdoor attacks. Backdoor attacks are a typical adversarial attack type on deep neural networks (DNNs). Gu et al. [3] first explored backdoor attacks, known as Badnets, for DNNs. During training, attackers embed hidden backdoor triggers in DNNs, enabling normal classification on benign samples. However, activating the hidden backdoor trigger causes the model to output the attacker's desired label. The backdoor attack was first introduced in FL by Bagdasaryan et al. [4]. The goal of the adversary is to plant backdoors in their local model such that the backdoored model will be aggregated into the global model, further inheriting the backdoor logic. Since the universal participation principle and invisible training procedure, backdoor attacks in FL emerge more destructive effects [5], [6], [7], [8], [9].

Several countermeasures [10], [11], [12], [13], [14], [15], [16] have been developed to alleviate the threats, which fall under the categories of either detection or mitigation methods (as shown in Table I). Detection methods [10], [12], [13] aim to utilize anomaly detection techniques to determine whether a model is backdoored. Although these methods typically show promising accuracies, the potential impact of backdoor triggers in the backdoored models is hard to eliminate. Furthermore, the effectiveness of detection methods is susceptible to the distribution of data in FL. Therefore, researchers are more focused on methods for backdoor mitigation [11], [14], [15], [16]. Mitigation methods in FL can be classified into three phases: pre-aggregation (Pre-Agg), in-aggregation (In-Agg), and post-aggregation (Post-Agg). Among these, In-Agg methods operate when the server aggregates local models, using techniques such as differential privacy, local validation [11],

Manuscript received 8 October 2023; revised 27 February 2024; accepted 27 March 2024. Date of publication 4 April 2024; date of current version 7 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62206238, Grant 62076125, Grant 62032025, and Grant U20B2049; in part by the Natural Science Foundation of Jiangsu Province under Grant BK20220562; in part by the Natural Science Research Project of Universities in Jiangsu Province under Grant 22KJB520010; in part by China Postdoctoral Science Foundation under Grant 2023M732985; in part by the State Key Laboratory of Massive Personalized Customization System and Technology under Grant H&C-MPC-2023-02-05; and in part by Shenzhen Science and Technology Program under Grant JCYJ20210324134810028. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Luca Cavaglione. (Jiale Zhang and Chengcheng Zhu are co-first authors.) (Corresponding authors: Chengcheng Zhu; Xiaobing Sun.)

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Digital Object Identifier 10.1109/TIFS.2024.3384846

TABLE I
COMPARISON OF STATE-OF-THE-ART BACKDOOR DEFENSES IN FL

Type	Method	Aggregation			Deployment Side	Defensive Assumption
		Pre	In	Post		
Detection	FoolsGold [10]	✓			Server	Non-IID
	Baffle [11]	✓			Server	Validation dataset
	DeepSight [12]	✓			Server	Non-IID
Mitigation	CONTRA [13]		✓		Server	Non-IID
	RLR [14]		✓		Server	-
	Pruning [15]			✓	Server	Validation dataset
	Unlearning [16]			✓	Server	Outsourced dataset
	<i>FLPurifier</i>	✓			Client	-

and robust learning rates [14] to mitigate the impact of backdoors. Post-Agg methods involve a protective process conducted after aggregating local models, potentially including malicious ones. The server takes a step further to amend the global model [15], [16], further purifying the adverse impacts on models caused by corrupt updates from malicious clients.

However, both In-Agg and Post-Agg methods are commonly executed on the server side and rely on some impractical assumptions such as clean labeled data or validation sets, which exhibit limited effectiveness and violate the intention of using FL to a large extent. Additionally, amending model parameters often causes considerable model performance degradation to achieve a high defense rate. More essentially, the aforementioned mitigation methods primarily focus on eliminating the strong associations between trigger features and target labels either in model aggregation or post aggregation. The exploration of this aspect leads to a fundamental yet far-overlooked question, “*Can we prevent the establishment of the connection between trigger features and the target label before federated aggregation?*”

Before answering this question, it is necessary to figure out how can the model learn a strong correlation between trigger features and the target label. As we all know, deep neural networks (DNNs) possess excellent capability of feature extraction. The training process of DNNs is substantially the learning process of feature, which shrinks the distance between the samples with the same label in the feature space. Such an end-to-end supervised learning paradigm makes the learned sample features strongly connect with their origin labels. However, the excessive feature extraction ability also leads to the backdoor trigger feature easily to be learned even if the trigger pattern is very stealthy or even invisible. The trigger feature will form an individual cluster connected to the target label under supervised training. *Therefore, if we can find a training mode that does not depend on labels, it is possible to break the clustering of backdoor features.*

A. Our Work

We propose *FLPurifier*, the first Pre-Agg backdoor mitigation method in FL. The key challenge of *FLPurifier* is to employ a local training mode to prevent the establishment of the connection between trigger features and the target label. To address the challenge, we introduce contrastive training into FL, which is an effective way to learn better data representation in a self-supervised manner. Fig. 1 illustrates how

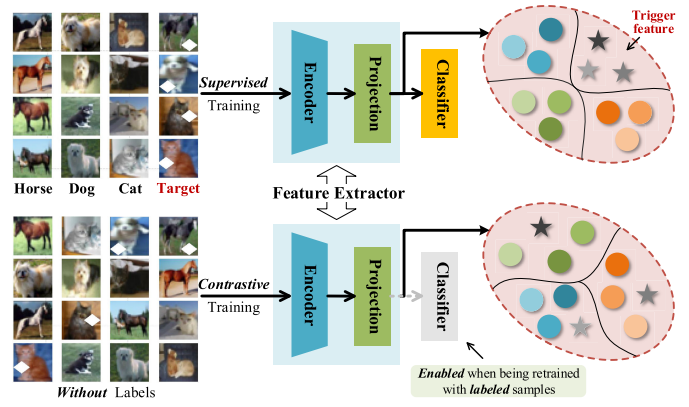


Fig. 1. Analysis of contrastive training against backdoor attacks.

contrastive training can defend against backdoor attacks in the local training procedure of FL. Intuitively, both of the networks can be regarded as feature extractors and classifiers. For contrastive training in FL, the trigger attributes are scattered among other cluster centers since the extractor is optimizing by contrasting the two augments of one sample rather than depending on the label of the sample. Moreover, *FLPurifier* utilizes adding noise or pixel blocks as data augmentation in contrastive training, which is similar to adding triggers to samples. However, these augmented samples are considered to be of the same class as the original samples during training instead of a target class. Such a way further destroys the clustering of backdoor attributes. Additionally, we further design an adaptive classifier aggregation to relieve the negative impact of malicious updates aggregated into the global extractor and leverage local data labels to improve the accuracy of the global model.

Our key contributions can be summarized as follows:

- We introduce a novel FL framework robust against backdoor attacks, called *FLPurifier*. Rather than mitigating the influence of backdoor in an already trained model, *FLPurifier* breaks the connection between trigger features and the target label in the training process by decoupled contrastive training.
- We design an adaptive classifier aggregation strategy to dynamically adjust the weight coefficients, further relieving the negative impact of backdoor classifiers and improving global model accuracy.
- We implement *FLPurifier* on six benchmark datasets. Experimental results show that *FLPurifier* can effectively reduce backdoor attack accuracy while maintaining high global model performance.

The remaining of this paper is organized as follows. Section II introduces the related works of backdoor attacks and defenses. The proposed method and algorithm are discussed in Section III. Section IV evaluates and analyses the results of the experiment. Finally, Section V concludes the paper.

II. RELATED WORK

A. Backdoor Attacks in Federated Learning

Backdoor attacks endeavor to manipulate the training data through the injection of adversarial triggers, causing the model

trained on the altered dataset to make erroneous predictions on the contrived samples. In contrast to the centralized learning framework, backdoor attacks in FL demonstrate heightened potency in causing damage due to the clandestine nature of the training process. In FL, backdoor attacks mainly contain two types, i.e., model replacement attacks [4], [17], [18] and distributed attacks [5], [6], [7]. For example, Bagdasaryan et al. first proposed the model replacement backdoor attack in FL by uploading the scaled-up local updates to replace the global model as a backdoored model [4]. Wang et al. [18] established that robustness to backdoors implies model robustness to adversarial examples and presented a strong backdoor attack method by utilizing edge-case adversarial training samples, forcing a model to misclassify on seemingly easy inputs. To augment the efficacy of backdoor attacks, Zhang et al. incorporated Generative Adversarial Networks (GANs) into the realm of such exploits [19]. Initially, the attacker emulates the prototypical samples derived from the training dataset of other participants and gains complete control over these generated samples, utilizing them to train the compromised models. Distinguished from the aforementioned centralized backdoor attacks, wherein each party embeds an identical global trigger during the training process, Xie et al. introduced the concept of distributed backdoor attack (DBA), which utilizes various independent local triggers to form a global trigger, and enhances the attack effect through the cooperation of attackers [5]. Afterward, various backdoor attacks have been investigated to increase the invisibility and robustness of triggers, such as inverting the input space to feature space or even latent representations [20], [21], selecting the poisoned samples that contribute more to the backdoor injection [22], and improving the persistence of backdoor features in the global model [23]. Besides, Rong et al. [24] have proved the feasibility of backdoor attacks on federated recommendation systems. In general, the backdoor triggers can be embedded in form of watermarking [25], pixel block [26], random noise [27], or attributes [4]. In this paper, we aim to defend against both model replacement and distributed backdoor attacks in FL that include all types of triggers mentioned above.

B. Backdoor Defense Solutions

For defending the aforementioned backdoor attacks in FL, existing defense methods mainly include two kinds: backdoor detection [11], [28] and backdoor mitigation [10], [13], [14], [29], [30]. Backdoor detection aims to identify whether there is a backdoor trigger in the target model, or filter suspicious samples directly from the training data for retraining [31], [32]. For example, Andreina et al. presented Baffle [11], a round-based feedback loop engaging clients in validating the global model, which exploits the availability of diverse datasets at the various clients by incorporating a feedback loop into the federated learning process, to integrate the views of those clients when deciding whether a given model update is genuine or not. However, these passive detection methods can only judge whether there is a backdoor attack in the model, whereas hard to eliminate the negative impact of the backdoor attack on the target model [33].

Lately, researchers began to explore how to purify the backdoored model by eliminating backdoor triggers. In centralized learning framework, the most common mitigation methods are mainly through fine-tune [34] or model-prune [35] on a subset of clean data, which will arise considerable model performance degradation. Besides, other methods such as model distillation [36] and machine unlearning [37] have also been proposed successively to reduce the effect of backdoor attacks. Specifically, Huang et al. [38] proposed to defend backdoor attacks by decoupling the original end-to-end training process, thus conducting a semi-supervised fine-tuning of the global model. Li et al. [39] aimed to break the connection between backdoor samples and the targeted class via the anti-backdoor learning. In the context of FL, Awan et al. introduced CONTRA [13], a cosine-similarity-based measurement method to evaluate the reliability of local model parameters in every iteration. Then, a reputation scheme is utilized to adjust the standing of individual clients dynamically, depending on their historical and per-round contributions to the global model. CRFL was proposed by Xie et al. [29], which exploits clipping and smoothing on parameters to keep the smoothness of the global model, yielding a sample-wise robustness certification on backdoors. Different from them, Ozdayi et al. [14] defend against backdoor attacks by adjusting the aggregation server's learning rate, per dimension and per round based on the sign information of the agent's updates, which is considered a lightweight defense since it just requires little change to federated learning protocol. In addition to these mitigation schemes, improving the robustness of the machine learning model against evasion and backdoor attacks has been well-devoted by researchers, such as RAB [40] and FLAME [41]. However, these methods either cause considerable model performance degradation or heavily rely on the assumption that the defender already owns plenty of labeled clean data whereas impractical in FL.

C. Contrastive Learning

Contrastive learning, a form of self-supervised learning, has garnered significant attention and implementation due to its remarkable performance [42]. Rooted in the field of deep learning, contrastive learning trains models by leveraging positive and negative sample pairs to enhance the model's ability to express features effectively. The central concept revolves around constructing pairs of similar instances, where these positive samples are brought close together in the feature space or projection space [43]. Conversely, dissimilar instances, forming negative sample pairs, are purposefully pushed far apart in the feature space. Contrastive learning has emerged as a promising method for learning better feature representation without supervision from labels. Many effective methods, e.g., SimCLR [44] and MoCO [45], rely on Siamese networks to minimize the similarity of two augmented views (positive pairs) and maximize the difference between two different images (negative pairs). Another line of work, e.g., BYOL [46] and SimSiam [47], even bypasses negative pairs and contrasts only positive pairs, employing stop-gradient operation to avoid trivial solutions.

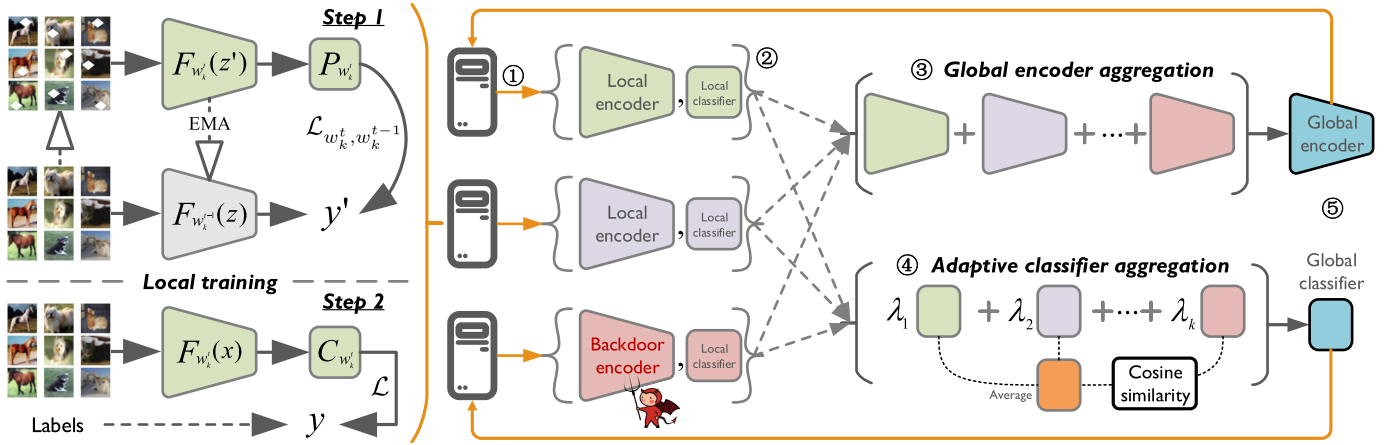


Fig. 2. **Workflow of *FLPurifier*:** ① *FLPurifier* initializes the global encoder and classifier and prompts clients to train them on local data. ② Local encoders and classifiers are computed via decoupled contrastive training. ③ Local encoders from all clients are aggregated at the server, which uses Fedavg. ④ Local classifiers from all clients are dynamically aggregated based on adaptive classifier aggregation strategy. ⑤ The global encoder and classifier are returned to the clients, who use them for local training. **Local Training:** Step 1: Taking the samples removed labels as input and employing the last round local encoder as the target model, the global encoder is trained via contrastive learning. Step 2: Combining the extractor with the classifier, the whole model is retrained on the labeled data in the manner of supervised learning.

Recently, several works that consider contrastive learning in FL are proposed to expand the unsupervised learning setting and handle the data heterogeneity [48], [49], [50], [51], [52]. For example, Li et al. proposed MOON [50], a model-level contrastive FL framework that incorporates the similarity between local model representations to correct the participants' training procedure, addressing the Non-IID issue in FL. Mendieta et al. demonstrated that the standard regularization method has shown surprising performance in eliminating data heterogeneity effects and meanwhile exhibits excellent resource efficiency. Based on this finding, a simple and effective method FedAlign [51] has been presented from the perspective of local learning generality to overcome the data heterogeneity. Although the previous study effectively addressed data heterogeneity in FL by using the powerful representation learning ability of contrastive training, it overlooked potential security vulnerabilities such as backdoor attacks. Therefore, in this paper, we add backdoor triggers as a mode of data augmentation on normal samples, and for the first time, contrastive training is applied to cut off the connection between the targeted label and the trigger feature, achieving the elimination of potential backdoor effects during the training phase.

III. FLPUFIER METHOD

A. Threat Model and Defense Goal

1) **Threat Model:** Given clean local samples D_k , the attacker aims to pollute a portion r of them by adding specific triggers Δ and tricks the model G to output desired target label c when taking the polluted samples as input. Meanwhile, the backdoor model can perform normally on clean samples. Thus, attackers manipulate the training process to minimize:

$$Loss = \sum_{(x,y) \in \mathcal{D}_k} \begin{cases} \mathcal{L}(G(x + \Delta), c) & r \\ \mathcal{L}(G(x), y) & 1 - r, \end{cases} \quad (1)$$

where $\mathcal{L}$ is the loss function. In FL setting, followed by recent backdoor attack methods [4], [18], we define the capabilities of attackers as follows: attackers can manipulate the process of the training phase where they can plant the trigger into the benign samples and assign them with the desired label. Then, they can upload the crafted parameters and replace the federated model with a backdoored model. Besides, the attacker has no access to interference with the server.

2) **Defense Goal:** For the abilities of defenders (i.e., benign participants), we define that they just need to comply with the established *FLPurifier* algorithm to execute the training procedure while no additional capabilities are required. Correspondingly, the defense goal of *FLPurifier* is to prevent the attacker from achieving its attack targets, which means the negative impact of backdoored model updates can be effectively eliminated by *FLPurifier* to significantly reduce the attack success rate (ASR). Besides, the main task performance (MA) of the global model must be guaranteed to ensure its utility.

B. Method

Fig. 2 presents the overview of our proposed framework, *FLPurifier*, designed for the backdoor defense in FL. *FLPurifier* instructs each client to execute a two-stage contrastive training to obtain a couple of separated feature extractor (encoder) $F_{w'_k}$ and classifier $C_{w'_k}$, which can effectively prevent the establishment of the connection between trigger features and the target label. Moreover, to further leverage the label information from each client and improve the global model accuracy, *FLPurifier* further introduces an adaptive classifier aggregation strategy. In the aggregation process, each client's local encoders are initially combined at the server using the Fedavg algorithm [1]. Subsequently, the proposed adaptive classifier aggregation strategy is applied to dynamically create the global classifier.

1) *Decoupled Contrastive Training*: As aforementioned, in the local training phase, each client executes supervised contrastive training, which consists of two stages. Specifically, at first stage, taking the local data whose labels are removed at this stage as input, contrastive learning with asymmetric Siamese networks (an online network and a target network) is executed. Adopted from BYOL [46], the online network consists of an online encoder and a predictor while the target network only contains a target encoder. Notably, different from the classifier used in the specific classification task, the predictor is used to output a regression target for the online network to contrast.

In FL, except for the first local training, we employ the global encoder $F_{\tilde{w}^t}$ and predictor $P_{w_k^t}$ as the local online encoder $F_{w_k^t}$ and predictor $P_{w_k^t}$ while the local target encoder is composed of the last round local online encoder $F_{w_k^{t-1}}$. Such settings make contributions to disperse the possible backdoor cluster in the global encoder aggregated by one or several backdoor encoders. That is, in the first training round, benign clients take the initial global encoder both as the online encoder and the target encoder, therefore, the trained encoder is clean and deserves confidence. The reliable encoder will be used as the target encoder in the next training round and purify the possible backdoor in the online encoder. In the following federated communication round, even though the backdoor encoder is aggregated into the global encoder again, we can utilize the last round local encoder to correct it. Formally, the contrastive loss can be defined as:

$$\begin{aligned} \mathcal{L}_{w_k^t, w_k^{t-1}} &\triangleq \left\| P_{w_k^t}(z) - F_{w_k^{t-1}}(z') \right\|_2^2 \\ &\triangleq 2 - 2 \cdot \frac{\left\langle P_{w_k^t}(z), F_{w_k^{t-1}}(z') \right\rangle}{\|P_{w_k^t}(z)\|_2 \cdot \|F_{w_k^{t-1}}(z')\|_2}, \end{aligned} \quad (2)$$

where $P_{w_k^t}(z)$ is the output of online predictor, $F_{w_k^{t-1}}(z')$ indicates the output of target encoder, t represents the communication round of FL, and k indexes the k -th participant. The online network and target network take two image augmentations results z and z' of the same image $x \sim D$ sampled uniformly from D_k as input, where the online network is updated with gradient descent and the parameters w_k^{t-1} of target network are an exponential moving average of the online network's parameters w_k^t after each training step:

$$w_k^{t-1} \leftarrow m w_k^{t-1} + (1 - m) w_k^t, \quad (3)$$

where $m \in [0, 1]$ is decay rate.

After the first stage, the local client trains a clean feature extractor by exploiting the information of the global feature extractor. To further realize the main task of FL and make the most of the label information each client possesses, in the second stage, we combine the extractor with the classifier and retrain the whole model on the labeled data in the manner of supervised learning. The objective function is as follows:

$$w_k^t = \arg \min_{w_k^t} \sum_{(x, y) \in \mathcal{D}_k} \text{CrossEntropy}(C_{w_k^t}(x), y). \quad (4)$$

2) *Adaptive Classifier Aggregation*: Each client sends the local encoder and classifier to the server for aggregation after local training. Then, *FLPurifier* performs the encoder aggregation and the classifier aggregation respectively. For the encoder aggregation, the sever averages the parameters based on the local data size to create the global encoder:

$$F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}, \quad (5)$$

where S_t denotes the set of the clients selected for the t -th FL and $F_{\tilde{w}^{t+1}}$ represents the global encoder for the next round.

For the classifier aggregation, considering that the classifier is trained by the labeled data to complete the specific classify task, while the attacker will train a classifier connecting the backdoor attribute with the target label. Although the backdoored classifier will not lead to misclassification when it is combined with the extractor since the extractor is obtained by contrastive learning and clean, the backdoored classifier may reduce the accuracy and break the performance of the global model. Therefore, *FLPurifier* introduces an adaptive classifier aggregation strategy to relieve the adverse effect caused by backdoored classifiers, further improving the accuracy of the main task. Specifically, we calculate the average classifier $\bar{C}_{w^t}$ reflecting the average level of all the clients according to the Eq. 6 at first. Then, the weight coefficient Q_k of each classifier is determined by the deviation degree between the local classifier and the average classifier. More precisely, the more the local classifier deviates from the average, the less its weight coefficient is. Generally, the backdoored classifier exists significant divergence from the normal classifier, thus its coefficient is relatively low, which leads to a limited influence on the global classifier. Finally, the server executes the adaptive classifier aggregation based on their coefficients:

$$\begin{cases} \bar{C}_{w^t} &\leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}, \\ Q_k &\leftarrow \frac{1}{\text{sim}(C_{w_k^t} \| \bar{C}_{w^t})}, \\ C_{\tilde{w}^{t+1}} &\leftarrow \sum_{k \in S_t} \lambda_k C_{w_k^t}, \end{cases} \quad (6)$$

where $\lambda_k = \frac{Q_k}{\sum_{k \in S_t} Q_k}$ is normalized weight coefficient of the k -th client, sim is cosine similarity used to quantify the degree of deviation between the two classifiers, and $C_{\tilde{w}^{t+1}}$ indicates the global classifier for the next round. The procedure of *FLPurifier* is presented in Algorithm 1, where the number of clients N and other parameters is taken as input and the global encoder $F_{\tilde{w}^T}$ and classifier $C_{\tilde{w}^T}$ are outputted as the final results.

IV. EXPERIMENTAL EVALUATION

To evaluate the effectiveness of our approach, we implemented our *FLPurifier* framework on Pytorch (1.8.0+cu111). All the experiments were conducted on a workstation with windows operating system, Intel i9-9700K CPU, 16GB memory, and NVIDIA GeForce

Algorithm 1 *FLPurifier* Algorithm

Input: number of clients N , number of communication round T , learning rate η , contrastive epoch E_{con} , supervised epoch E_{sup} .

Output: The final global encoder $F_{\tilde{w}^T}$ and classifier $C_{\tilde{w}^T}$.

- 1 **Server executes:**
- 2 Initialize the global encoder, predictor and classifier $F_{\tilde{w}^0}, P_{\tilde{w}^0}, C_{\tilde{w}^0}$;
- 3 **for** each communication round $t \in [1, 2, \dots, T]$ **do**
- 4 $S_t \leftarrow$ (random set of N clients);
- 5 **for** each client $k \in S_t$ **do**
- 6 $F_{w_k^t}, C_{w_k^t} \leftarrow$ **LocalTraining**($k, F_{\tilde{w}^t}, C_{\tilde{w}^t}$);
- 7 // Encoder, predictor aggregation
- 8 $F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}$;
- 9 // Adaptive classifier aggregation
- 10 $\tilde{C}_{w^t} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}$;
- 11 **for** each client $k \in S_t$ **do**
- 12 $Q_k \leftarrow \frac{1}{sim(C_{w_k^t} || \tilde{C}_{w^t})}$;
- 13 $C_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{Q_k}{\sum_{k \in S_t} Q_k} C_{w_k^t}$;
- 14 **Return** $F_{\tilde{w}^T}, G_{\tilde{w}^T}$;
- 15 **Client executes:**
- 16 **LocalTraining**($k, F_{\tilde{w}^t}, C_{\tilde{w}^t}$) ;
- 17 $F_{w_k^t}, C_{w_k^t} \leftarrow F_{\tilde{w}^t}, C_{\tilde{w}^t}$;
- 18 // Contrastive training for Extractor
- 19 **for** each local epoch $k \in [1, 2, \dots, E_{con}]$ **do**
- 20 **for** each batch $b = \{x\} \in D_k$ **do**
- 21 $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{w_k^t, w_k^{t-1}}(w_k^t; x)$;
- 22 $w_k^{t-1} \leftarrow m w_k^{t-1} + (1 - m) w_k^t$;
- 23 // Supervised training for Classifier
- 24 **for** each local epoch $k \in [1, 2, \dots, E_{sup}]$ **do**
- 25 **for** each batch $b = \{x, y\} \in D_k$ **do**
- 26 $\mathcal{L}_{sup} \leftarrow CrossEntropy(C_{w_k^t}(x), y)$;
- 27 $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{sup}$;
- 28 **Return** $F_{w_k^t}, C_{w_k^t}$.

GTX2080 GPU. In this section, we designed comprehensive experiments to answer the following three research questions.

- **RQ1** (Superiority of *FLPurifier*): What are the advantages of *FLPurifier* compared with state-of-the-art (SOTA) methods?
- **RQ2** (Effectiveness of *FLPurifier*): How can *FLPurifier* achieve backdoor defense during training phase?
- **RQ3** (Applicability of *FLPurifier*): Does *FLPurifier* still be effective in different FL settings?
- **RQ4** (Overhead of *FLPurifier*): How does the communication and computational cost of *FLPurifier* perform?

A. Datasets and Experimental Setup

1) *Datasets*: *FLPurifier* is evaluated on six widely-used benchmark datasets: MNIST, Fashion-MNIST (F-MNIST), EMNIST, CIFAR-10, GTSRB, and CIFAR-100. Note that EMNIST is used for Non-IID settings. For the encoder, *FLPurifier* is compatible with various model architectures. We consider several popular models: CNN model for MNIST and F-MNIST, VGG for CIFAR-10, and DenseNet for GTSRB. For the CIFAR-100 which is a 100-classes dataset, we employ Resnet-18.

a) *Training configurations*: In our experiments, federated training protocol is run with 100 clients (with averaged training samples) where 30 of them are malicious. The attackers embed a trigger in the training images and assign it with a target label and all the attackers construct the backdoor trigger in the same way for a consistent target. During the training procedure, each participant uses SGD as optimizer and trains for E local epochs with local learning rate η , and a batch size of 64. For the local training process, we set the attacker's local epochs and initial learning rate as $E = 20$ and $\eta = 0.05$, while the benign participants are $E = 10$ and $\eta = 0.1$. Besides, 10 participants (three of which are malicious) are randomly selected in each round to submit local updates for aggregation. Such settings make contribution to the success of backdoor attacks. Note that above settings are used for backdoor attacks based on different types of triggers while we follow original settings for the baseline attacks.

b) *Evaluation metrics*: We evaluate the performance of defense mechanisms with two metrics: attack success rate (ASR), which is the ratio of backdoored samples misclassified as the labels specified by attackers, and the accuracy of the main classification task on normal samples (MA). A robust FL framework means significantly reducing the ASR while maintaining a high MA.

c) *Baselines*: We test the effect of the defense methods against three kinds of known backdoor attacks in FL, i.e., CAS [4], DBA [5], and Edge-case (PGD) [18]. Moreover, we also choose four typical backdoor defense methods in FL as our comparison baselines, i.e., FoolsGold [10], CONTRA [13], Baffle [11], and RLR [14].

B. Comparison With SOTA Methods (RQ1)

Table II shows the performance of *FLPurifier* compared with four baseline backdoor defense methods, all these methods using the same settings as in the original works. For backdoor attacks, we set the scaling factor $S = 30$ in CAS attack, and other attacks follow their original setting. Besides, column "Before" is listed as the original baseline without any defense and the best results are in Bold.

The experimental show that *FLPurifier* can lower the ASR of all four backdoor attacks to 4% on the four benchmark datasets, which surpasses the performance of the other three defense methods. Overall, all defense methods in the experiments are effective against centralized backdoor attacks while FoolsGold fails to defend against DBA which is in a distributed manner. Specifically, CONTRA and Baffle have a similar level in the defense of backdoor attacks while the effect

TABLE II
PERFORMANCE COMPARISON BETWEEN *FLPurifier* AND FOUR SOTA DEFENSE METHODS

Datasets	Attack	Before		FoolsGold		CONTRA		Baffle		RLR		<i>FLPurifier</i>	
		ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA
MNIST	DBA	100	90.43	97.73	79.23	7.13	87.45	8.64	82.97	6.51	86.19	2.69	90.32
	Edge-case	92.47	93.01	12.59	81.51	5.34	88.13	7.43	84.06	2.37	88.52	1.47	92.57
	PGD	90.39	91.12	11.87	82.05	4.29	88.26	3.26	85.72	1.94	85.88	0.94	92.05
F-MNIST	DBA	100	89.67	98.69	80.05	6.79	86.92	9.05	84.39	5.64	84.47	2.31	90.19
	Edge-case	91.58	92.56	10.52	81.20	2.85	87.48	3.47	85.83	3.05	85.93	0.85	91.44
	PGD	89.76	92.03	11.62	79.28	3.27	88.03	4.71	83.85	2.12	87.04	1.26	91.89
CIFAR-10	CAS	79.24	87.21	9.15	80.16	3.44	82.79	3.58	79.41	8.85	82.58	1.63	88.14
	DBA	94.12	72.14	92.42	71.57	8.51	71.55	9.82	70.56	7.92	69.87	3.24	75.66
	Edge-case	58.61	83.56	8.31	81.89	4.39	80.97	5.95	79.13	5.91	80.92	1.32	82.43
	PGD	68.35	79.43	7.94	78.50	5.36	76.19	6.70	77.32	5.14	73.65	1.58	79.21
GTSRB	CS	81.77	89.53	10.58	80.21	5.98	87.20	7.93	83.39	5.06	85.78	2.17	89.11
	DBA	100	89.37	98.95	79.61	6.80	86.82	7.11	82.35	5.87	87.85	2.50	89.92
	Edge-Case	87.32	92.59	10.64	81.40	3.16	85.30	4.56	84.44	1.78	86.28	1.79	90.87
	PGD	90.26	91.93	11.61	79.76	3.69	86.10	4.77	82.20	1.45	85.84	1.56	89.27
CIFAR-100	CS	76.32	61.36	8.84	55.31	3.10	56.41	3.97	54.85	3.49	58.09	2.07	62.02
	DBA	86.91	61.95	92.54	54.33	8.38	57.15	7.55	53.58	5.86	56.68	2.81	62.36
	Edge-Case	53.25	63.01	8.35	56.58	4.02	55.45	5.26	53.44	5.49	55.11	0.96	61.65
	PGD	59.17	61.65	8.14	57.00	5.39	55.96	3.94	54.63	3.69	57.90	1.41	60.18

of FoolsGold is slightly inferior. As the backdoor mitigation method, RLR relieving backdoor with robust learning rate shows better performance in defending backdoor against in FL. However, all of the baseline defense methods cause degeneration in ACC while *FLPurifier* exhibits a superior defense effect and maintains model performance degradation at 2%, which is a distinct superiority of *FLPurifier*. Surprisingly, some values of MA from *FLPurifier* are even higher than the “Before” column. To our analysis, it is thanks to the adaptive classifier aggregation that can also reduce the weight of low-quality classifiers.

C. Further Understanding of *FLPurifier* (RQ2)

1) *Robustness Under Different Triggers*: To further verify the robustness of *FLPurifier* To further verify the robustness of *FLPurifier* to different trigger types and sizes, we select four types of triggers: watermarking, pixel blocks, random noise, and imperceptible input. For the watermarking attack, 0 is applied as the watermarking and the coefficient is varied from 0.1 to 0.5, where their span is 0.2. In the backdoor attacks based on pixel blocks, we extend the trigger sizes from 3×3 to 5×5 and 7×7 . Besides, for the random noise attacks, we change the noise tensy from 10 to 20 and 30. All aforementioned backdoor triggers used in our experiments are shown in Fig. 3.

Table III, IV, and V summarize the performance comparison of *FLPurifier* with other backdoor defense baselines. Overall, all defense methods are nearly immune to the three trigger types and sizes above. In more detail, Baffle detects abnormal updates and CONTRA adjusts the weight of malicious clients. For RLR, it relieves the backdoor with a robust learning rate. The bigger sizes and higher intensity seem to improve the ASR without any backdoor defense. It is because the feature of the trigger is harder with the increase of the trigger sizes,

which leads to the feature being easier to be learned by the model. For different types of triggers, the ASR of the imperceptible input is significantly greater than the other three types of triggers. However, after the defenses are deployed, the backdoored updates are hard to be aggregated into the global model, or their weight is lowered, no matter the trigger types and sizes. For the *FLPurifier*, we directly cut off the connection between trigger features and the target label in the training process, which is also independent of the trigger type. The above findings also indicate that the trigger size and type, which are significant factors in centralized backdoor research, hardly affect the performance of defenses in FL.

In addition, we partly visualize and compare the confusion matrix without and with *FLPurifier* in Fig. 4 under the attack whose triggers are pixel blocks. In MNIST task, the test samples in which the original labels are “0” were embedded in a 4×4 pattern and assigned with a target label “8”. For CIFAR-10 task, we respectively assign the 4-th class (“deer”) with the label of the 7-th class (“horse”) under the same setting as MNIST. Intuitively, results indicate that most of the samples “7” with triggers are misclassified into “4”. The same situation occurs in CIFAR-10 dataset. However, the vast majority of crafted inputs are classified into their original labels with the help of *FLPurifier*.

Moreover, we visualize the distribution in feature space of the global extractor under the above attacks with/without *FLPurifier* to further understand the advantages of our proposed method. More specifically, we poisoned 10% of the test data in CIFAR-10, where the number of each poisoned class is the same. Then, the poisoned test data are used for t-SNE visualization. Fig. 5 including six subfigures shows the t-SNE vitalization. From Fig. 5a to Fig. 5b, we can see that the feature representations of benign samples from the same category form an individual cluster, while the poisoned samples form a new cluster (in black color). However, the clusters of

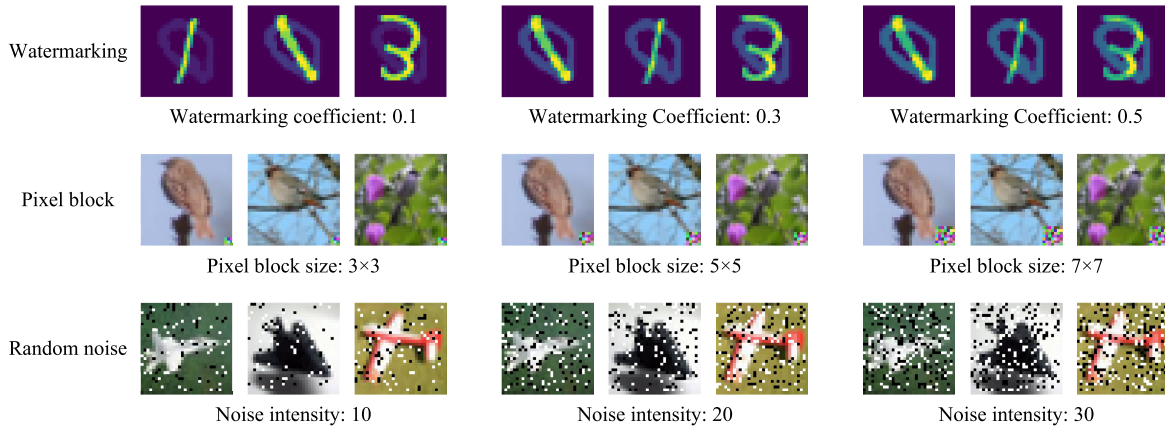


Fig. 3. Examples of backdoored images by different trigger types and size.

TABLE III
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON MNIST

Backdoor attacks	Trigger size	Before		FoolsGold		CONTRA		Baffle		RLR		FLPurifier	
		ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA
Imperceptible input		93.78	91.32	18.87	81.35	13.42	84.27	15.92	81.41	6.62	86.42	2.92	90.55
Watermarking	0.1	89.06	91.58	11.02	82.23	4.02	87.45	5.38	85.43	3.03	88.24	1.24	91.41
	0.3	91.25	92.83	12.75	81.51	3.68	88.13	4.75	84.35	2.84	88.01	1.49	93.01
	0.5	92.87	92.88	10.72	82.05	4.36	88.26	4.86	85.65	2.59	87.79	2.05	90.88
Pixel block	3x3	90.44	90.89	10.44	82.05	3.52	86.92	3.74	84.51	3.12	88.25	0.89	89.96
	5x5	92.18	93.54	9.32	81.20	3.44	87.48	3.92	86.51	3.13	88.57	1.35	93.20
	7x7	90.11	90.89	10.02	82.28	2.12	88.03	3.73	85.90	1.66	87.32	1.26	90.17
Random noise	10	89.48	92.60	11.43	81.16	3.73	87.79	4.19	85.50	2.57	86.71	1.63	89.60
	20	90.01	93.29	14.93	82.57	4.10	87.55	4.53	85.49	3.44	87.89	2.24	91.23
	30	91.20	92.79	12.25	81.89	2.94	86.97	4.78	85.20	3.39	87.62	2.32	91.77

TABLE IV
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON CIFAR-10

Backdoor attacks	Trigger size	Before		FoolsGold		CONTRA		Baffle		RLR		FLPurifier	
		ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA
Imperceptible input		92.79	86.83	18.64	79.32	14.20	79.55	16.71	80.46	5.43	82.19	2.92	85.97
Watermarking	0.1	78.53	84.43	7.82	79.65	3.57	82.04	4.06	81.97	2.74	81.59	1.25	84.56
	0.3	79.76	83.59	8.63	79.32	3.70	81.35	5.62	81.06	3.36	82.52	1.09	83.25
	0.5	78.86	84.35	6.82	80.03	3.96	82.21	4.82	80.72	3.34	83.01	1.36	83.98
Pixel block	3x3	76.54	85.42	8.17	79.45	2.90	82.44	3.11	79.39	2.55	82.42	0.97	83.06
	5x5	76.19	83.23	7.71	78.94	3.21	83.17	4.06	81.83	2.79	83.04	1.26	83.54
	7x7	76.55	84.43	9.77	78.95	2.53	82.19	3.86	80.85	2.74	82.14	0.74	82.71
Random noise	10	77.50	87.16	7.14	78.62	3.42	83.75	4.68	82.41	2.77	82.47	1.31	84.04
	20	76.85	85.85	8.67	79.37	4.18	82.61	6.70	81.56	3.35	81.60	2.39	85.11
	30	76.23	84.09	7.51	79.18	4.68	83.13	5.67	82.13	3.99	82.51	1.48	83.54

poisoned samples in Fig. 5e to Fig. 5f are absolutely damaged, all the poisoned samples afresh assemble with the benign samples from the same category. This means that *FLPurifier* can effectively break the cluster of backdoor features and further prevent the establishment of the connection between the targeted label and the trigger through decoupled contrastive training. Moreover, the benign samples locate in their own cluster in a relatively concentrated manner, where the distance between samples within clusters is very close meanwhile there is a significant distance between clusters, illustrating that the extractor shows excellent ability on the main classify task.

2) *Adaptive Classifier Aggregation*: To further explore the impact of adaptive classifier aggregation (ACA), we process more experiments with/without ACA on IID and Non-IID settings, respectively. For the scene without ACA, we adopt Fedavg algorithm to aggregate the classifiers. Table VI and VII represent the result of the experience. In general, both the defense effect and the main task accuracy have a degeneration after we remove the ACA. In specific, the MA decreased by about 6% at most, which means that the ACA makes significant contributions to the maintenance of the MA. According to our analysis, it is because ACA first calculates the averaged

TABLE V
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON CIFAR-100

Backdoor attacks	Trigger size	Before		FoolsGold		CONTRA		Baffle		RLR		FLPurifier	
		ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA
Imperceptible input		89.40	61.27	30.88	51.16	7.50	57.79	9.29	55.41	2.49	57.58	1.33	60.65
Watermarking	0.1	77.07	62.30	9.59	51.23	3.08	57.45	4.55	55.74	2.82	57.19	2.31	61.67
	0.3	78.30	63.54	11.89	51.51	4.32	57.13	4.59	55.18	3.32	56.52	2.47	62.45
	0.5	79.42	62.60	10.95	52.05	3.68	57.26	3.62	55.61	2.93	57.88	1.94	62.71
Pixel block	3x3	73.95	62.19	9.07	52.05	3.82	58.92	2.07	57.28	2.28	58.47	0.93	62.12
	5x5	75.72	62.45	9.79	52.20	3.11	57.48	3.53	57.01	2.27	57.93	0.96	62.02
	7x7	76.48	61.12	8.80	53.28	3.66	57.03	3.99	57.80	1.93	57.04	0.76	60.17
Random noise	10	79.40	61.27	11.88	51.16	3.50	57.79	4.29	55.39	2.49	57.58	1.33	60.65
	30	80.48	62.07	13.18	50.57	4.52	58.55	5.48	54.56	3.27	56.87	2.24	62.12
	50	81.92	61.32	11.60	51.89	4.56	56.97	4.34	53.15	3.33	58.02	2.37	59.97

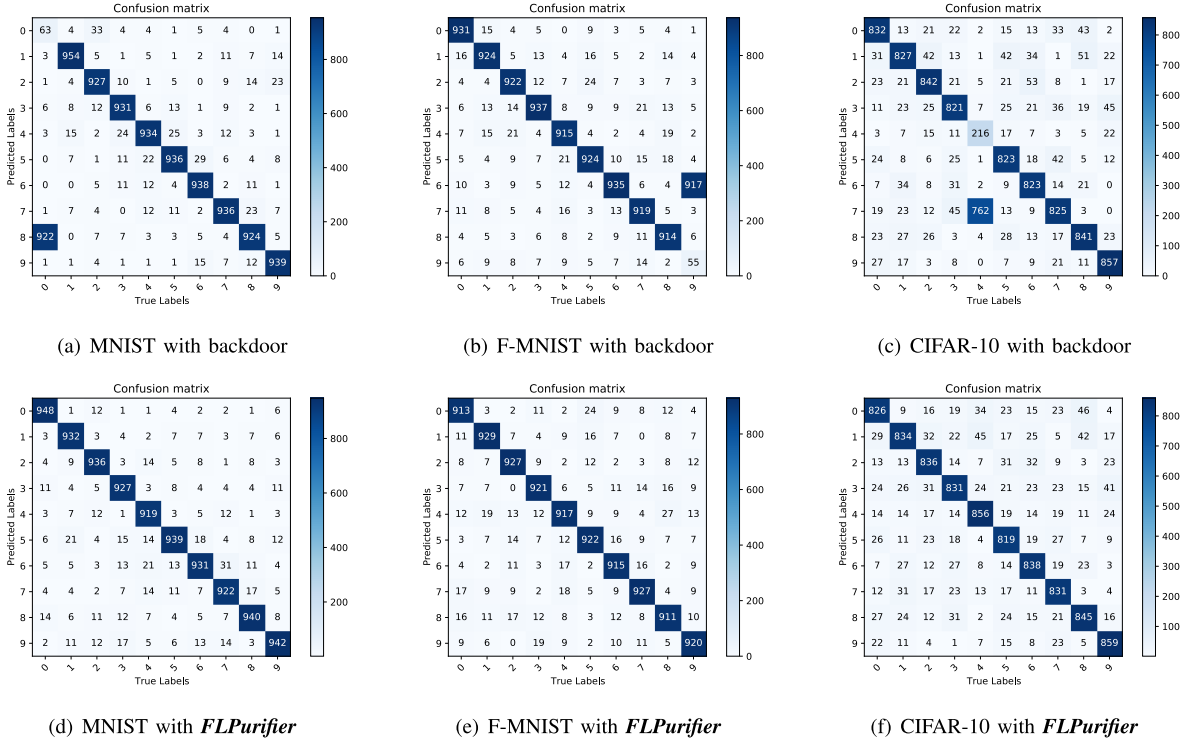


Fig. 4. Confusion matrix of the proposed *FLPurifier* and baseline backdoor model on MNIST, F-MNIST, and CIFAR-10 datasets.

weight classifier which represents the universal trend of classification, and then dynamically adjusts the weight according to the deviation between each classifier and it. Such settings reduce the weight of low-quality classifiers resulting from the limitation of the quality of the local data, and also reduce the weight of the backdoor classifier since the backdoor classifier also has a distance from the average classifier. In contrast, the increase in ASR is slight because the success of backdoor defense is mainly due that we break the connection between trigger features and the target label in the training process by decoupled contrastive training while the ACA just plays an auxiliary role. Moreover, comparing the IID and Non-IID settings, we can draw that ACA achieves better performance meaning that ACA is more needed in Non-IID settings since the data of each client in Non-IID settings are unbalanced and thus the local classifier trained on these data are divergent.

ACA can relieve these problems and improve the consistency of the classifiers.

3) *Different Ratios of Malicious Client*: To ensure the effectiveness and reliability of FL in the presence of backdoor attacks, it is crucial for defense methods to be robust to varying proportions of malicious participants. In this part, we aimed to assess the robustness of different defense methods against backdoor attacks at varying malicious participation ratios. Specifically, we set the malicious participation ratio r which represents the ratio of the number of malicious clients appearing in each round of FL to the total number of clients selected in that round. We investigated the MA and ASR of *FLPurifier* compared with four defense methods under three types of attacks. Note that the curve of Fedavg represents the result without any defense. Fig. 6 illustrates six subfigures of the trend of MA (the upper) and ASR (the lower)

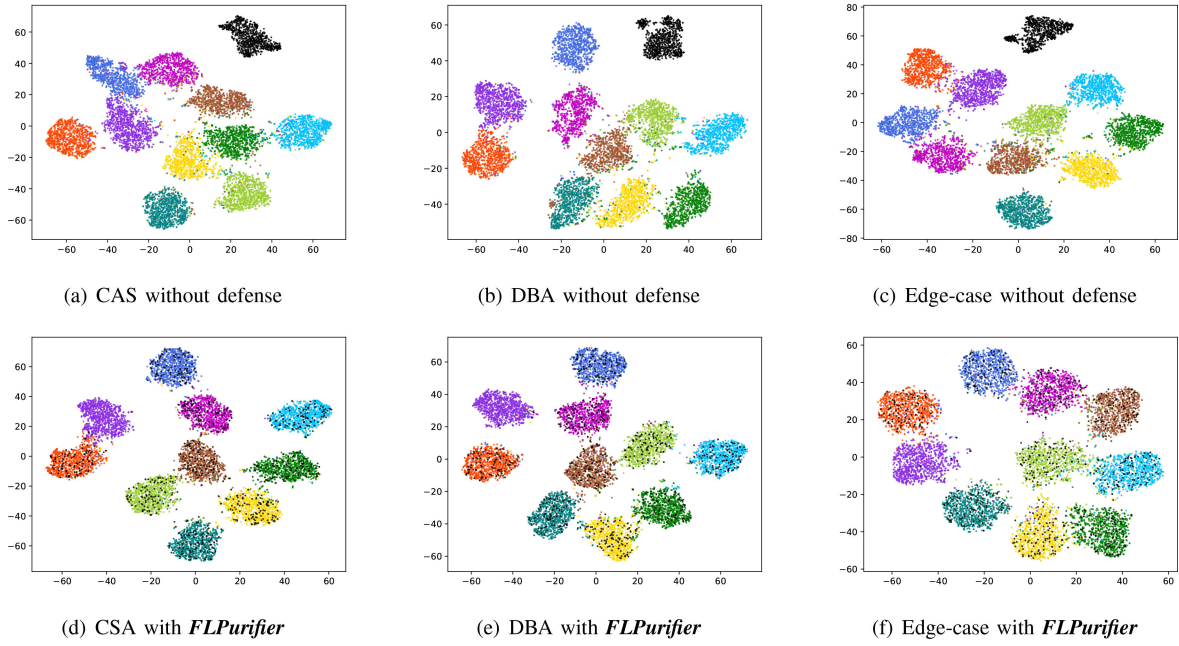


Fig. 5. The t-SNE of the feature representations under different attacks on CIFAR-10 *with/without* *FLPurifier*.

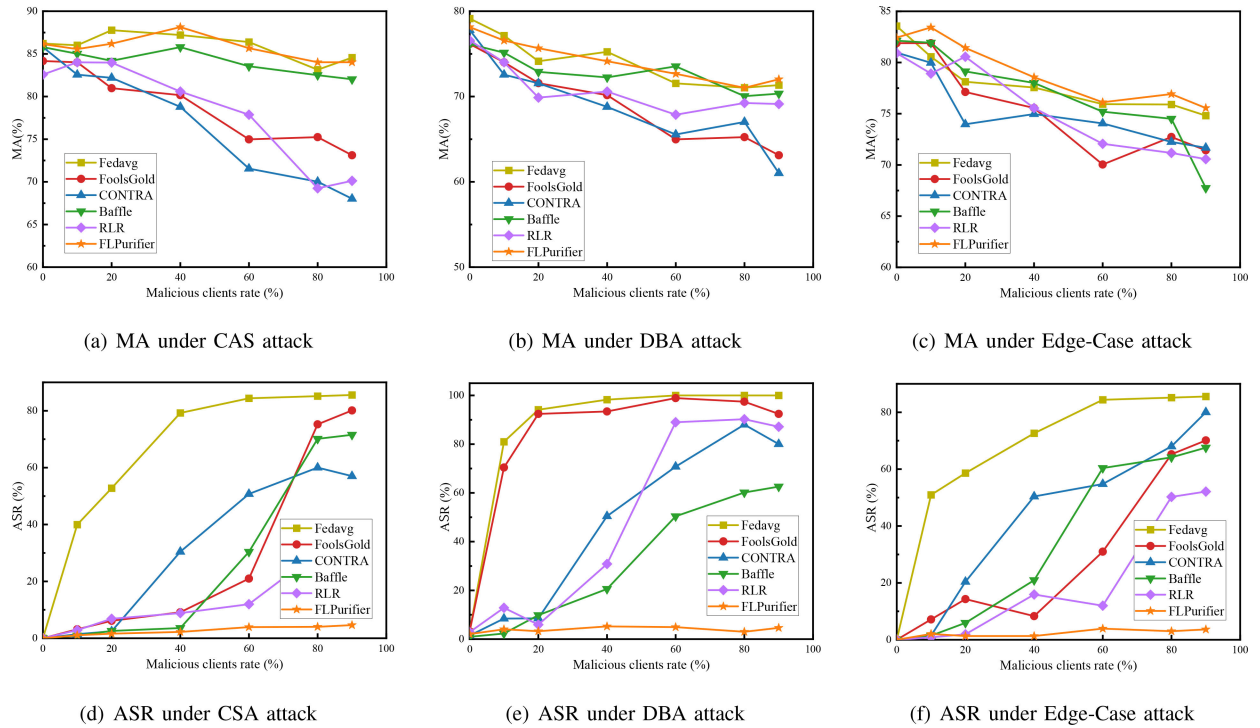


Fig. 6. The defense performance of *FLPurifier* against CAS, DBA, and Edge-Case attacks with the different rates of malicious clients on CIFAR-10 dataset.

with the increasing malicious participation ratio for a specific attack.

As we can observe that the MA shows a decreasing trend as the malicious participation ratio increase while the ASR appears a significant increase. It is reasonable because more malicious participants mean more backdoors are embedded, causing greater disturbance to the global model. Meanwhile, the increasing of malicious clients implies plenty of backdoor data are used to train the target model, which causes improvement in ASR. However, *FLPurifier* maintained relatively high

main task accuracy and effectively mitigated the success rate of backdoor attacks across different malicious participation ratios. On the contrary, baseline methods face various challenges at higher malicious participation ratios. FoolsGold defended the backdoor by reducing the weight of malicious participants. However, when the number of attackers is sufficient, even if their weight is reduced, it can still promote the success of backdoor attacks. Similarly, CONTRA implements a measure through a reputation scheme to determine the credibility of local model parameters to dynamically adjust the

TABLE VI

COMPARISON RESULTS OF DEFENSE PERFORMANCE WITH OR WITHOUT ACA ON IID SETTING

Dataset	Attack	Without ACA		With ACA	
		ASR	MA	ASR	MA
MNIST	DBA	4.73	85.14	2.69	90.32
	Edge-Case	1.59	89.21	1.47	92.57
	PGD	1.08	87.32	0.94	92.05
F-MNIST	DBA	4.76	86.16	2.31	90.19
	Edge-Case	2.91	87.72	0.85	91.44
	PGD	1.96	88.14	1.26	91.89
CIFAR-10	CS	3.13	84.91	1.63	88.14
	DBA	4.75	70.59	3.24	75.66
	Edge-Case	2.12	80.56	1.32	82.43
GTSRB	PGD	3.02	76.54	1.58	79.21
	CS	1.98	87.23	2.17	89.11
	DBA	3.16	74.65	2.50	89.92
CIFAR-100	Edge-Case	1.82	83.74	1.79	90.87
	PGD	3.61	74.98	1.56	89.27
	CS	3.03	60.17	2.07	62.02
CIFAR-100	DBA	3.74	56.13	2.81	62.36
	Edge-Case	1.17	57.46	0.96	61.65
	PGD	2.09	54.54	1.41	60.18

TABLE VII

COMPARISON RESULTS OF DEFENSE PERFORMANCE WITH OR WITHOUT ACA ON NON-IID SETTING

Dataset	Setting	Without ACA		With ACA	
		ASR	MA	ASR	MA
MNIST	$\alpha = 0.05$	4.78	78.12	1.59	80.97
	$\alpha = 0.1$	6.04	81.77	3.04	86.72
	$\alpha = 10$	4.73	85.14	2.69	90.32
E-MNIST	$\alpha = 0.05$	6.94	59.95	3.21	66.37
	$\alpha = 0.1$	5.98	63.32	3.56	69.68
	$\alpha = 10$	3.57	71.09	2.81	74.96

weight of each client, and the reputation scheme is ineffective under excessive malicious clients. Battle verifies whether there is a backdoor in the model through the private data of the client, and decides whether to abandon it through a vote strategy. Correspondingly, RLR determined the local update sign through voting. Obviously, when there are enough malicious participants, these mechanisms relying on voting are actually fatal. Instead, *FLPurifier* eliminates backdoor attacks from the root by decoupling the training process of encoders and classifiers, thereby being hardly affected in ASR by the ratios of malicious clients. However, the massive malicious clients cause limitations to ACA, which drops the MA slightly. Overall, the experiment results highlight the robustness of federated learning to different ratios of malicious clients.

4) *Different Attack Interval*: In certain scenarios, attackers may be randomly selected in each round, or the frequency of attacks may be reduced to enhance their concealment. Thus, we investigate the performance of *FLPurifier* under reduced or even zero attack frequencies. Specifically, we conduct experiments with different attack intervals to evaluate *FLPurifier* against DBA on the CIFAR-10 dataset. The experimental results, presented in Table VIII, demonstrate that

TABLE VIII

PERFORMANCE OF *FLPurifier* UNDER DIFFERENT ATTACK INTERVAL

Attack Interval	Before		<i>FLPurifier</i>	
	MA	ASR	MA	ASR
Non-attack	-	-	84.57	-
0	72.14	94.12	75.66 ($\uparrow 3.23$)	3.24 ($\downarrow 90.88$)
1	72.04	93.75	75.98 ($\uparrow 3.27$)	3.27 ($\downarrow 90.48$)
2	77.59	87.93	78.42 ($\uparrow 3.01$)	3.01 ($\downarrow 84.92$)
4	80.25	82.48	82.75 ($\uparrow 2.56$)	2.56 ($\downarrow 79.92$)
8	81.47	76.62	83.14 ($\uparrow 1.79$)	1.79 ($\downarrow 74.83$)
16	83.01	69.86	84.07 ($\uparrow 1.21$)	1.21 ($\downarrow 68.65$)

DBA maintains a relatively high attack success rate, despite a moderate decrease in performance with reduced attack frequencies. Furthermore, as the attack interval increases, the damage inflicted by DBA on MA is mitigated. As expected, *FLPurifier* also shows improved defense capabilities with lower attack frequencies. It is worth noting that even when deployed without any attacks, *FLPurifier* can maintain a high level of MA. These findings indicate that the attack frequency has no significant impact on the performance of *FLPurifier*, and the deployment of *FLPurifier* does not inflict harm on the federated learning process.

D. Applicability Analysis (RQ3)

1) *Number of Clients*: The number of clients is a key setting in the FL scenarios. To verify the applicability of our proposed method, we evaluate the impact on the performance of *FLPurifier* by varying the number of clients. Moreover, we extend the scaling factor S from 10 to 50 in the CAS attack and do experiments on six datasets to better evaluate the robustness of *FLPurifier*. Note that since MNIST, F-MNIST, and EMNIST are hard to find some nature feature to execute CAS backdoor attack, we set the scaling factor following the setting in CAS but evaluate against the DBA attack. In general, *FLPurifier* is effective in different client number settings. Although we can see from Fig. 7 that the success rate of backdoor attacks decreases with the increase in the number of clients, we think it is mainly because more clients decrease the aggregation weights of attackers. Additionally, *FLPurifier* is hardly affected by attack intensity since the amplified weight of the backdoor classifier is effectively alleviated by the adaptive aggregation strategy. This is because as the scaling factor increases, the difference between the backdoor classifier and the average classifier also amplifies, which is easily detected by ACA, leading to a decrease in the weight of the backdoor classifier.

2) *Non-IID Data Distributions*: The different Non-IID data distributions are another significant and realistic setting in the FL scenarios where data is distributed in Non-IID fashion among agents. Following prior art [53], we model Non-IID data distributions using a Dirichlet distribution $\text{Dir}(\alpha)$, where a smaller α indicates higher data heterogeneity. Specifically, we vary α from 0.05 to 0.1 and 10 on MNIST and EMNIST against DBA, where $\alpha = 10$ represents IID. Table IX reports the final results for the Non-IID setting. The results indicate that *FLPurifier* provides an effective defense with negligible

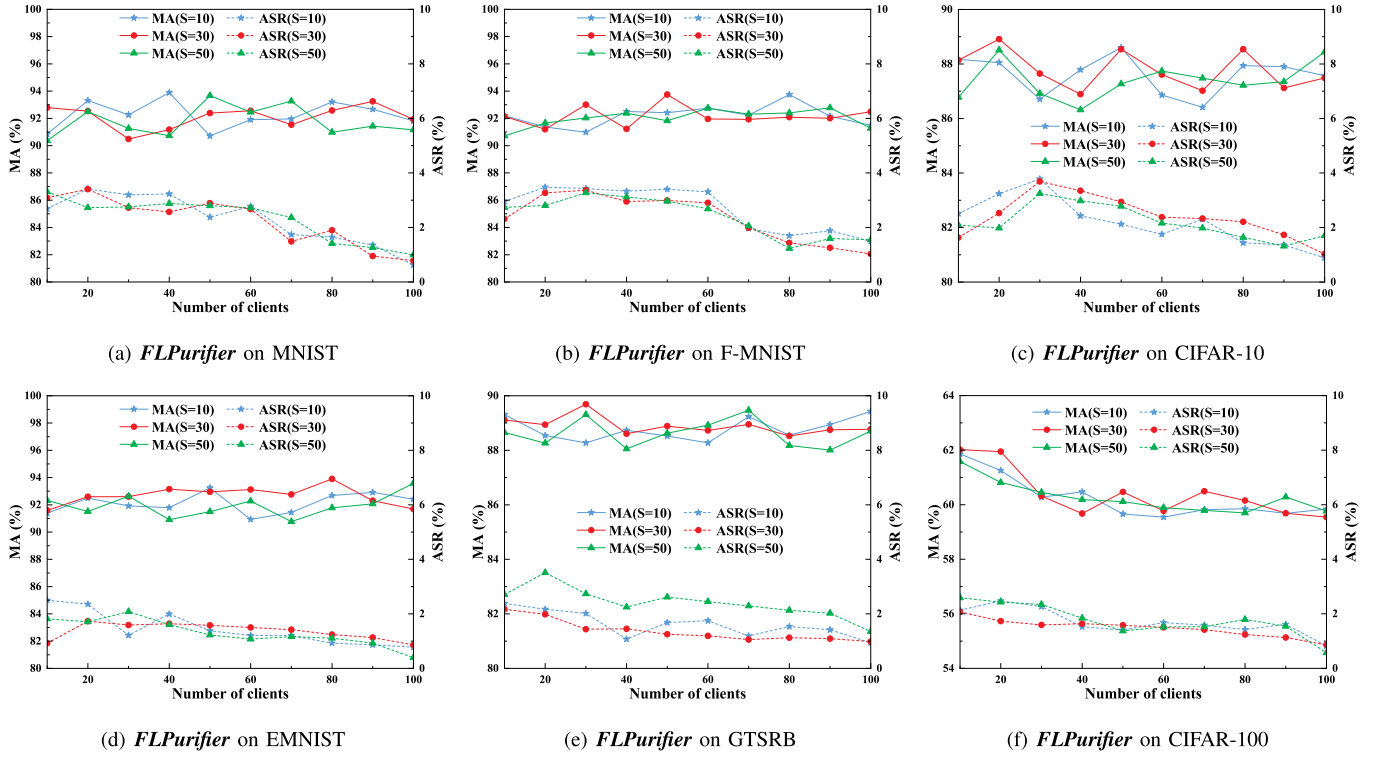


Fig. 7. The defense performance of *FLPurifier* against CAS attack with the different numbers of clients on six datasets.

degradation in the main task accuracy which outperforms other methods. In more detail, *FLPurifier* lowers the ASR of DBA to below 4 % under the Non-IID setting, while the baseline methods exhibit evident degradation, where only the RLR still retains unsatisfactory results (round 9%). Surprisingly, FoolsGold which is proved particularly effective in Non-IID has almost lost its effect in defending against DBA. For CONTRA, it measures the alignment level among clients and a higher alignment level means that the client's local objective is closer to the one of another client, which is more suspicious to be malicious. However, the alignment level is hard to measure since the data distribution is extremely imbalanced. Similarly, the imbalanced data distribution also enhances the difficulty of Battle in detecting malicious updates. In contrast, *FLPurifier* demonstrated excellent results by mitigating backdoors during the training process. Therefore, it can be drawn that *FLPurifier* is almost independent of data distribution and has excellent applicability.

3) *Adaptive Attacks*: As previously discussed, the success of backdoor attacks stems from the injection process where the attacker adds triggers to a normal sample and modifies its label. These trigger features form a distinct cluster associated with the target label under supervised training. To address this issue, our proposed approach involves decoupling the encoder and classifier structures of the model through contrastive training, thereby breaking the connection between backdoor features and target labels. Assume that the attacker is aware of our defense intuition and method. A feasible approach for them is to implant backdoor attacks without relying on attack-specific labels. In other words, they would employ untargeted or clean-label backdoor attacks. In this scenario, the attacker

directly adds triggers to a clean sample while maintaining its original label (e.g., “dog”). During the training phase, the attacker expects the model to learn the trigger features present in the sample and associate that the trigger belongs to the dog class. Consequently, in the prediction stage, the sample containing embedded triggers would be misclassified as “dog.” The fundamental nature of these attacks lies in establishing a connection between the trigger features and the normal features of the target class (the trigger features and the normal features of the target class form a cohesive cluster in the feature space), rather than establishing a direct connection between the trigger features and the target label (where the trigger features act as an individual cluster center). This idea aligns with previous invisible and clean-label backdoor attacks, thus we adopt this method as our adaptive attacker strategy.

As shown in Table X, *FLPurifier* still exhibits high performance by significantly reducing ASR to a mere 3% across all datasets. Furthermore, it maintains a negligible decrease in MA and even demonstrates slight improvements on complex datasets. These outcomes can be primarily attributed to the trick employed during the local training process, which is detailed in the methods section. In brief, *FLPurifier* leverages the local encoder of the last federated communication round as the target encoder while employing the global encoder as the online encoder. Despite adaptive attacks embedding a backdoor into the online encoder, resulting that the outputs of the online encoder for the samples with triggers are similar to that for the same samples without triggers due to the formation of a cohesive cluster between trigger features and normal features of the samples in the feature space. However, the target encoder lacks this characteristic. Consequently, the

TABLE IX
COMPARISON RESULTS OF DIFFERENT FEDERATED BACKDOOR DEFENSE METHODS ON MNIST AND EMNIST (NON-IID)

Datasets	Setting	Before		FoolsGold		CONTRA		Baffle		RLR		<i>FLPurifier</i>	
		ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA	ASR	MA
MNIST	$\alpha = 0.05$	84.52	84.09	83.26	76.92	39.74	80.89	42.18	79.21	11.56	80.14	1.59	83.95
	$\alpha = 0.1$	90.63	87.85	89.43	78.81	24.56	82.62	28.16	80.17	9.15	81.59	3.04	86.83
	$\alpha = 10$	100	90.43	97.73	79.23	7.13	87.45	8.64	82.97	6.51	86.19	2.69	90.32
EMNIST	$\alpha = 0.05$	80.71	67.96	78.74	63.49	36.72	61.49	39.12	62.73	9.11	62.12	3.21	67.54
	$\alpha = 0.1$	85.67	72.42	80.55	67.66	35.31	64.53	40.12	64.16	9.24	63.39	3.56	73.05
	$\alpha = 10$	94.29	76.78	87.12	70.09	17.62	71.02	11.53	71.42	7.87	72.10	2.81	75.18

TABLE X
PERFORMANCE OF *FLPurifier* UNDER ADAPTIVE BACKDOOR ATTACK

Attack	Dataset	Before		<i>FLPurifier</i>	
		ASR	MA	ASR	MA
Adaptive	MNIST	96.3	92.8	2.6	91.6
	F-MNIST	95.8	91.9	3.1	89.5
	EMNIST	94.2	75.7	1.7	75.4
	CIFAR-10	97.1	83.7	0.9	84.1
	GTSRB	96.9	89.5	2.5	88.3
	CIFAR-100	91.8	59.7	1.4	60.8

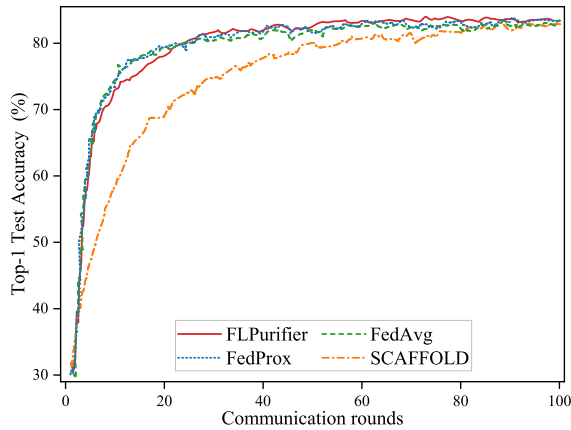


Fig. 8. The defense performance of *FLPurifier* compared with different aggregation methods on CIFAR-10 dataset.

target encoder exerts a corrective effect on the online encoder. Upon completing the training, the backdoor features are effectively eliminated. These findings affirm the effectiveness of our method in countering adaptive attacks.

E. Overhead of *FLPurifier* (RQ4)

Figure 8 illustrates the accuracy achieved in each round during training by comparing *FLPurifier* with three SOTA FL frameworks, i.e., Fedavg [1], FedProx [54], and SCAFFOLD [55]. Note that no attackers were included in these experiments. According to Figure 8, we can observe that the speed of accuracy improvement in *FLPurifier* is slightly lower compared with FedAvg and FedProx in the initial rounds (before 20 rounds). However, they achieve convergence and even higher accuracy in nearly the same number of rounds. This observation suggests that *FLPurifier* does not incur additional communication overhead compared to traditional

TABLE XI
COMMUNICATION AND COMPUTATION COST OF *FLPurifier*

Target Accuracy	Communication (rounds)				Computation (epochs)			
	E=2	E=6	E=10	E=20	E=2	E=6	E=10	E=20
70%	60	30	15	8	120	180	150	160
80%	70	45	30	20	140	270	300	400

federated learning algorithms. However, during the server aggregation phase, there is a computational cost associated with calculating the similarity between different local models. This overhead, although present, is deemed acceptable considering the need for maintaining security in federated learning settings, particularly when mobile phones, which often serve as clients in FL, require robust security guarantees but possess limited computational resources. On the other hand, the server typically possesses substantial computational capabilities, making it better suited to handle such overhead efficiently.

Table XI presents a comprehensive analysis of the necessary communication rounds and computation epochs for achieving a target accuracy using different local epochs E which is the sum of E_{con} and E_{sup} . Increasing E reduces the communication cost as it decreases the required rounds, but it generally leads to higher computation costs. For example, consider the scenario where $E = 6$, which necessitates 45 rounds of communication to achieve 80% accuracy with 270 computation epochs. On the other hand, if we set $E = 20$, only 8 rounds of communication are required, but it demands 400 computation epochs. These findings highlight the inherent trade-off between the communication cost and the computation cost in the context of federated learning.

V. SUMMARY AND FUTURE WORK

In this paper, we proposed *FLPurifier*, a novel backdoor defense framework in FL based on supervised contrastive training. Specifically, *FLPurifier* instructs clients to separately train a clean feature extractor by contrastive learning and a classifier in a supervising manner, thus breaking the connection between backdoor features and target labels. To further eliminate the impact of malicious classifiers and improve the accuracy of the global model, we present an adaptive classifier aggregation strategy, dynamically adjusting the weight coefficients of each classifier according to their deviation degree. Compared with existing defense methods, *FLPurifier*

can eliminate the negative impacts of backdoor attack in the training procedure. The extensive experiments validate that *FLPurifier* is effective against backdoor attacks and outperforms the existing defense. In future work, we will investigate how to deploy *FLPurifier* in the scenario of heterogeneous or personalized federated learning.

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